12TH CHEMISTRY · CHAPTER 23 · FULL CHAPTER NOTES

Carbonyl Compounds and Carboxylic Acids

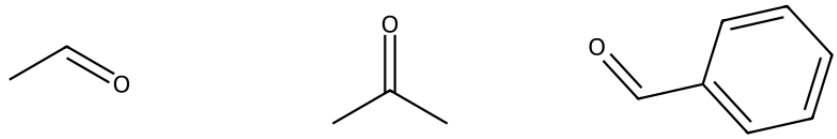
Topic-wise short and long questions with answers from the complete chapter

23.1 CARBONYL COMPOUNDS

Q.1 [Short] What is the carbonyl group? What are aldehydes and ketones? Where are they found and used?

In organic chemistry, the structural unit >C=O is known as the **carbonyl group**. Aldehydes and ketones are the classes of compounds that contain a carbonyl group. Their general structures are R–CHO (aldehydes) and R–CO–R (ketones).

These compounds are abundant in nature and large quantities are synthesised every year. Aldehydes are present in many spices, fruits and natural fragrances. Ketones are used in the pharmaceutical, chemical and paint industries. The most common ketone is acetone (propanone), which is found in nail polish removers.



ethanal (an aldehyde) propanone (a ketone) benzaldehyde

Q.2 [Short] What gives bitter almonds their smell and taste?

The smell and taste of bitter almond is due to benzaldehyde, which is produced by the oxidation of terpenes.

Q.3 [Short] What is formalin? Give its use.

Methanal, HCHO, is a gas at room temperature. It is often encountered as a 40 % aqueous solution called **formalin**, which has a fishy, unpleasant smell and is used for preserving biological specimens. Other short-chain aldehydes and ketones are liquids.

The textbook box prints “Methanol, HCHO”; HCHO is methanal.

Q.4 [Long] How are aldehydes prepared? Why is the aldehyde distilled off as soon as it is formed?

Aldehydes are prepared by the oxidation of primary alcohols with acidified $K_2Cr_2O_7$ or acidified $KMnO_4$.

$$CH_3–CH_2–OH + [O] \rightarrow CH_3–CHO + H_2O \ (K_2Cr_2O_7 + H_2SO_{4(aq)}, \text{heat})$$

In the preparation of ethanal, the acidified $K_2Cr_2O_7$ is added dropwise to warm ethanol. The aldehyde is **distilled off as soon as it is formed**; this prevents its further oxidation to ethanoic acid. The temperature is kept above the boiling point of ethanal but below that of ethanol, since ethanol has the higher boiling point. Pure ethanal is obtained by redistillation.

Q.5 [Short] In the preparation of ethanal, why is the temperature kept between the boiling points of ethanal and ethanol?

Ethanal has a lower boiling point than ethanol. At a temperature above the boiling point of ethanal but below that of ethanol, the ethanal distils off as soon as it is formed, while the unreacted ethanol stays in the flask. This prevents further oxidation of ethanal to ethanoic acid.

Q.6 [Short] How are ketones prepared? Why is the ketone not distilled off immediately?

Ketones are prepared by the oxidation of secondary alcohols, by the same procedure. Since ketones are not converted into carboxylic acids under these conditions, the ketone is not distilled off immediately.

$$CH_3\text{–}CH(OH)\text{–}CH_3 + [O] \rightarrow CH_3\text{–}CO\text{–}CH_3 + H_2O \ (K_2Cr_2O_7 + H_2SO_4, \text{heat})$$

23.2 CHEMISTRY OF CARBONYL COMPOUNDS

Q.7 [Short] Name the two basic types of reactions of carbonyl compounds.

The carbonyl group governs the chemistry of aldehydes and ketones. There are two basic reactions of carbonyl compounds:

- A. Nucleophilic addition reactions
- B. Alpha (α) carbon specific reactions

Q.8 [Long] Describe the structure of the carbonyl group. Which features decide its reactivity?

In the carbonyl group both carbon and oxygen are sp^2 hybridized. Carbon is bonded to the oxygen atom through a double bond and to two other groups through single bonds. The group is polar, because oxygen, being more electronegative, gets a partial negative charge (δ–) and carbon gets a partial positive charge (δ+). Two structural features determine the chemistry of aldehydes and ketones:

1. The carbonyl carbon is electrophilic in nature due to its δ+ charge, and can be attacked by nucleophiles.
2. The double bond can undergo addition reactions because the π bond is easily broken down. Moreover, the planar shape of the group makes the carbonyl carbon uncrowded, giving further support to addition reactions.

Q.9 [Short] Why is the carbonyl group polar?

Oxygen is more electronegative than carbon, so it attracts the shared electrons and gets a partial negative charge (δ–), while carbon gets a partial positive charge (δ+).

Q.10 [Long] Why are aldehydes more reactive than ketones towards nucleophiles?

Aldehydes are more reactive than ketones for two reasons:

1. **Inductive effect:** in a ketone, the electron-releasing (inductive) effect of two alkyl groups reduces the positive charge on the carbonyl carbon, making it less easily attacked by nucleophiles.
2. **Steric hindrance:** the two alkyl groups of a ketone tend to get in the way of the attacking nucleophile.

Reactivity towards nucleophiles: $H_2C{=}O > RHC{=}O > R_2C{=}O$

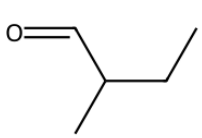
Q.11 [Short] What is steric hindrance in ketones?

The two alkyl groups attached to the carbonyl carbon of a ketone tend to get in the way of the attacking nucleophile. This crowding is called steric hindrance; it increases from methanal to aldehydes to ketones and lowers the reactivity.

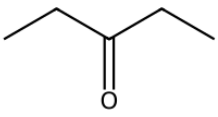
Quick Check 23.1

Q.12 [Short] a) Draw the structures for the following according to IUPAC system. i) 2-,methylbutanal ii) pentan-3-one iii) 3-methylbutenal d) 4-methylpentan-3-one

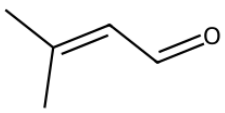
	Name	Condensed structure
i)	2-methylbutanal	$CH_3–CH_2–CH(CH_3)–CHO$
ii)	pentan-3-one	$CH_3–CH_2–CO–CH_2–CH_3$
iii)	3-methylbutenal (3-methylbut-2-enal)	$(CH_3)_2C{=}CH–CHO$
d)	4-methylpentan-3-one	$CH_3–CH_2–CO–CH(CH_3)_2$



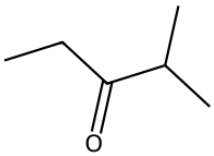
i) 2-methylbutanal



ii) pentan-3-one



iii) 3-methylbut-2-enal

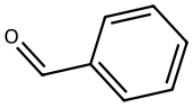


d) 2-methylpentan-3-one

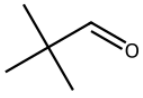
The fourth part is lettered d) in the textbook. The name "4-methylpentan-3-one" does not use the lowest numbers; the structure it describes is correctly named 2-methylpentan-3-one.

Q.13 [Short] b) Name the following compounds (IUPAC) system). i) C_6H_5–CHO ii) $(CH_3)_3C$–CHO iii) $(CH_3CH_2CH_2)_2C{=}O$

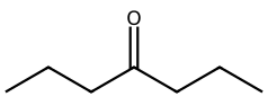
i) benzaldehyde (benzenecarbaldehyde) **ii)** 2,2-dimethylpropanal **iii)** heptan-4-one



i) benzaldehyde



ii) 2,2-dimethylpropanal



iii) heptan-4-one

Q.14 [Short] **c) Rank the following compounds in order of increasing reactivity towards nucleophilic attack. i) $(CH_3)_3CCOCH_3$ ii) $(CH_3)_3CCOC(CH_3)_3$ iii) $(CH_3)_2CO$**

$(CH_3)_3CCOC(CH_3)_3$ (ii) < $(CH_3)_3CCOCH_3$ (i) < $(CH_3)_2CO$ (iii)

The bulkier and the more electron-releasing the alkyl groups on the carbonyl carbon, the greater the steric hindrance and the smaller its positive charge, so the lower the reactivity.

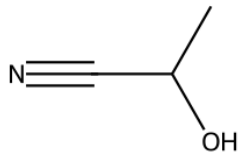
23.3 REACTIONS OF CARBONYL COMPOUNDS — Nucleophilic addition

Q.15 [Long] **Describe the addition of hydrogen cyanide to aldehydes and ketones.**

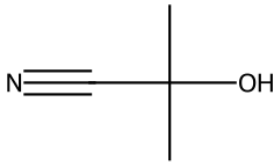
Aldehydes and ketones react with hydrogen cyanide to form cyanohydrins (hydroxynitriles). HCN itself is not used directly, as it is an extremely poisonous gas. Instead, it is produced by slowly adding a mineral acid (HCl) to a mixture of aqueous sodium cyanide or potassium cyanide and the carbonyl compound.

CH_3–CHO + HCN → CH_3–CH(OH)–CN (KCN/HCl) 2-hydroxypropanenitrile

CH_3–CO–CH_3 + HCN → $(CH_3)_2C(OH)$–CN (KCN/HCl)
2-hydroxy-2-methylpropanenitrile



2-hydroxypropanenitrile (from ethanal)



2-hydroxy-2-methylpropanenitrile (from propanone)

Q.16 [Short] **Why is HCN not used directly in the preparation of cyanohydrins?**

HCN is an extremely poisonous gas. It is produced in the reaction mixture itself by slowly adding a mineral acid (HCl) to a mixture of aqueous NaCN or KCN and the carbonyl compound.

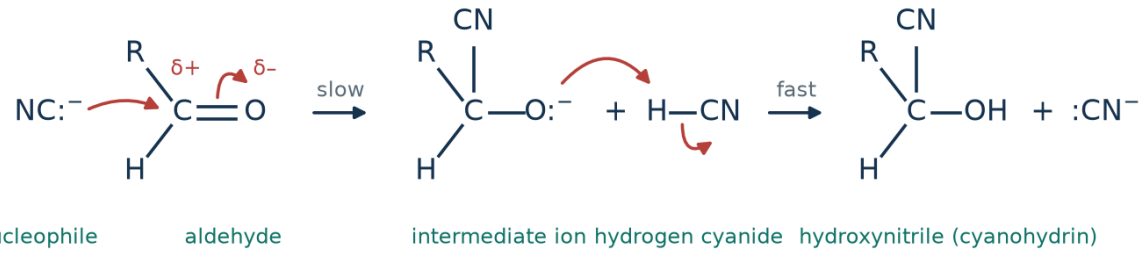
Q.17 [Long] **Explain the mechanism of the nucleophilic addition of HCN to carbonyl compounds.**

HCN is neither a nucleophile (there is no lone pair on its carbon atom) nor a strong acid that ionises to a reasonable extent, so it does not produce enough nucleophile (CN^-) to start the reaction. Therefore a slight excess of KCN is used, so that some free CN^- ions are present. These ions start the reaction and catalyse it.

i) KCN (excess) + HCl (limiting) → HCN + KCl

ii) The nucleophile CN^- attacks the δ+ carbonyl carbon; the π electrons move to the oxygen, giving an intermediate ion with a negative charge on oxygen.

iii) This ion takes a proton from HCN to form the hydroxyalkanenitrile, and a CN^- ion is regenerated.



Although both steps are reversible, the equilibria lie far to the right, so most carbonyl compounds give high yields of cyanohydrins.

Q.18 [Short] Why is a slight excess of KCN used in the reaction of HCN with carbonyl compounds?

HCN is neither a nucleophile nor a strong acid, so it does not give enough CN^- ions to start the reaction. With a slight excess of KCN, some free CN^- ions are present; they start the reaction and catalyse it so that it moves at a reasonable rate.

Q.19 [Short] What happens when nitriles are hydrolysed?

Nitriles are hydrolysed by dilute acid or aqueous alkali to form carboxylic acids or carboxylate ions.

Q.20 [Short] What is the industrial use of the hydroxynitrile obtained from propanone?

The hydroxynitrile obtained from propanone (2-hydroxy-2-methylpropanenitrile) is used as an intermediate in the manufacture of Perspex, a very fine coloured-glass polymer.

The textbook box names this compound "2-hydroxy-2-ethylpropanenitrile"; the product it draws from propanone is 2-hydroxy-2-methylpropanenitrile.

Q.21 [Long] Describe the reduction of aldehydes and ketones (addition of hydride ion).

Since aldehydes and ketones are synthesised by the oxidation of alcohols, alcohols can be produced by their reduction. With a reducing agent such as $NaBH_4$ or $LiAlH_4$, aldehydes give primary alcohols and ketones give secondary alcohols.

$$CH_3–CHO + 2[H] \rightarrow CH_3–CH_2–OH \ (NaBH_4, \text{heat; } H_2O/OH^-)$$

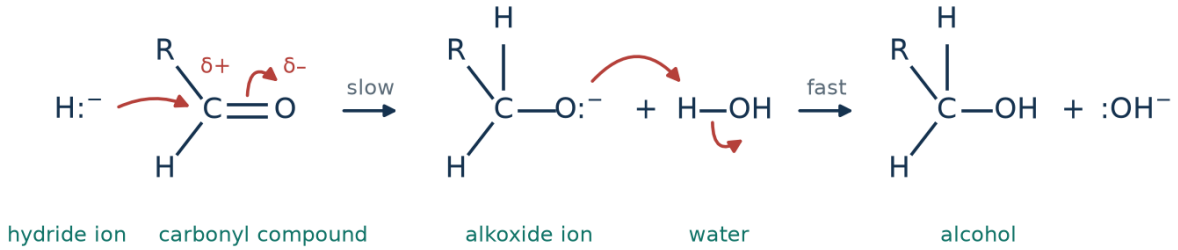
$$CH_3–CO–CH_3 + 2[H] \rightarrow CH_3–CH(OH)–CH_3 \ (NaBH_4, \text{heat; alkaline aqueous solution})$$

Q.22 [Short] What does [H] represent in the equations of reduction?

The reducing hydrogen in brackets, [H], comes from $NaBH_4$ and is used to balance the equation. It does not represent atomic hydrogen.

Q.23 [Long] Give the mechanism of the reduction of a carbonyl compound by the hydride ion.

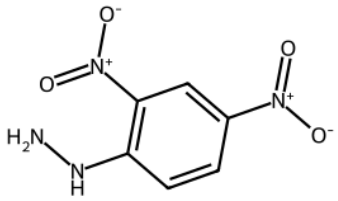
The reducing agent supplies the hydride ion, H^-, which is the nucleophile. It attacks the δ+ carbonyl carbon and the π electrons shift to oxygen, forming an **alkoxide ion**. The alkoxide ion then takes a proton from water to give the **alcohol** and a hydroxide ion.



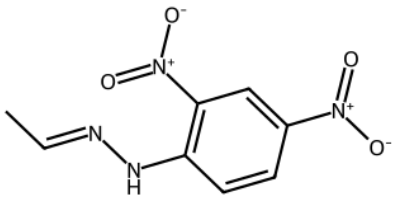
Q.24 [Long] Describe the reaction of carbonyl compounds with 2,4-dinitrophenylhydrazine. What is its use?

Aldehydes and ketones undergo a condensation (addition–elimination) reaction with 2,4-dinitrophenylhydrazine (2,4-DNPH) to give orange crystalline solids with well-defined melting points. A molecule of water is eliminated.

$$CH_3–CHO + H_2N–NH–C_6H_3(NO_2)_2 \rightarrow CH_3–CH=N–NH–C_6H_3(NO_2)_2 + H_2O$$



2,4-dinitrophenylhydrazine



ethanal 2,4-dinitrophenylhydrazone

This reagent is used to identify carbonyl compounds; for example, ethanal gives orange crystals of ethanal 2,4-dinitrophenylhydrazone. Other classes of compounds which also contain the >C=O group, such as carboxylic acids, esters, acyl halides and amides, do not give this test.

The reagent is abbreviated "2,4-NDPH" in the text of the book and "2,4-DNPH" in the learning outcomes and figure caption; 2,4-DNPH is correct.

Q.25 [Short] Why is the 2,4-DNPH reaction called a condensation (addition–elimination) reaction?

The reagent first adds to the carbonyl group and then a small molecule, water, is eliminated, forming a C=N bond. The joining of two molecules with the loss of a small molecule is condensation.

Q.26 [Short] Do carboxylic acids and esters give the 2,4-DNPH test?

No. Although carboxylic acids, esters, acyl halides and amides also contain the >C=O group, they do not give this test. Only aldehydes and ketones give the orange crystalline solid.

23.3.2 TEST OF ALDEHYDES AND KETONES

Q.27 [Short] Why are aldehydes easily oxidised while ketones are not?

Aldehydes and ketones are oxidised to carboxylic acids. Aldehydes are easily oxidised, whereas ketones are reluctant towards oxidation because a C–C bond has to be broken during oxidation. Therefore ketones are oxidised only by strong oxidising agents, whereas aldehydes are oxidised by both weak and strong oxidising agents.

Q.28 [Long] What is Tollens' test? How does it distinguish aldehydes from ketones?

Tollens' reagent, also called ammoniacal silver nitrate solution, contains silver ions in aqueous alkaline solution. These silver ions are readily reduced to silver metal on gentle warming with an aldehyde, and a **silver mirror** is formed along the wall of the test tube. Ketones do not give this test.

$$AgNO_3 + 3NH_4OH \rightarrow [Ag(NH_3)_2]OH + NH_4NO_3 + 2H_2O$$

Q.29 [Short] How was the silver mirror reaction used for making domestic mirrors?

A dilute solution of glucose (an aldehydic sugar) was used to make mirrors for domestic use by means of this reaction.

Q.30 [Long] What is Fehling's solution test?

Fehling's solution is a bright blue solution of Cu^{2+} ions in an aqueous alkaline medium, complexed with a salt of tartaric acid. When warmed with an aldehyde, the Cu^{2+} ions are reduced to Cu^{+} ions, which in the alkaline solution form a **brick-red precipitate** of copper(I) oxide. Ketones do not give this test.

$$R\text{–}CHO + 2Cu(OH)_2 + NaOH \rightarrow R\text{–}COONa + Cu_2O\downarrow + 3H_2O$$

Q.31 [Long] What is the iodoform test? Which compounds give it?

Aldehydes and ketones readily undergo substitution by halogens at the alpha position. Among aldehydes only ethanal, and among ketones only methyl ketones, possess the group CH_3CO–, known as the acetyl (ethanoyl) group. These compounds react with iodine in the presence of sodium hydroxide to form **triiodomethane (iodoform)**, a yellow crystalline solid. Hence the appearance of a yellow precipitate identifies the CH_3CO– group in an unknown carbonyl compound.

$$CH_3\text{–}CHO + 3I_2 + 4NaOH \rightarrow CHI_3\downarrow + HCOONa + 3NaI + 3H_2O$$

$$CH_3\text{–}CO\text{–}CH_3 + 3I_2 + 4NaOH \rightarrow CHI_3\downarrow + CH_3COONa + 3NaI + 3H_2O$$

In the propanone equation the textbook prints the salt as HCOONa; with propanone the salt formed is sodium ethanoate, CH_3COONa, which also balances the equation.

Q.32 [Short] Why does ethanol give the iodoform test while methanol does not?

Alcohols having the CH_3–CH(OH)– group also give the iodoform test. Ethanol gives the test, while methanol does not: ethanol has the CH_3–CH_2–O structure, which acts like a methyl ketone structure (it is oxidised to CH_3–C=O), so it can form iodoform. Methanol has no such group.

$$CH_3CH_2OH + 4I_2 + 6NaOH \rightarrow CHI_3 + HCOONa + 5NaI + 5H_2O$$

Q.33 [Short] Which aldehyde gives the iodoform test, and why only this one?

Only ethanal, CH_3–CHO, because it is the only aldehyde having the CH_3CO– (acetyl) group.

Q.34 [Long] Summarise the tests used to identify aldehydes and ketones.

Test	Reagent	Positive result	Given by
2,4-DNPH	2,4-dinitrophenylhydrazine	Orange crystalline solid	Aldehydes and ketones
Tollens' test	Ammoniacal silver nitrate	Silver mirror	Aldehydes only
Fehling's test	Alkaline Cu^{2+} with tartrate	Brick-red precipitate of Cu_2O	Aldehydes only
Iodoform test	I_2 and NaOH	Yellow precipitate of CHI_3	Compounds with CH_3CO– or $CH_3CH(OH)$–

Quick Check 23.2

Q.35 [Short] a) If compound A has the molecular formula C_3H_6O, it reacts with $NaBH_4$ to give compound B, C_3H_8O. Both A and B give a yellow precipitate with I_2/NaOH. Identify both compounds (A & B).

A gives the iodoform test, so it contains the CH_3CO– group: **A is propanone, CH_3–CO–CH_3**. Its reduction gives a secondary alcohol with the CH_3–CH(OH)– group, which also gives the test: **B is propan-2-ol, CH_3–CH(OH)–CH_3**.

Q.36 [Short] b) Why methanol does not give iodoform test while ethanol does?

Alcohols having the CH_3–CH(OH)– group also give the iodoform test. Ethanol gives the test, while methanol does not: ethanol has the CH_3–CH_2–O structure, which acts like a methyl ketone structure (it is oxidised to CH_3–C=O), so it can form iodoform. Methanol has no such group.

Q.37 [Short] **c) Describe a test (a different one in each case) that you could use to distinguish between the following pairs of compounds. i) CH_3CH_2CHO and CH_3COCH_3 ii) $CH_3CH_2COCH_2CH_3$ and $CH_3CH_2CH_2COCH_3$ iii) CH_2=CH–CH_2–OH and CH_3CH_2CHO**

i) Tollens' test: warm each with ammoniacal silver nitrate. Propanal (an aldehyde) gives a silver mirror; propanone does not.

ii) Iodoform test: warm each with I_2 and NaOH. Pentan-2-one (a methyl ketone) gives a yellow precipitate of CHI_3; pentan-3-one does not.

iii) 2,4-DNPH test: propanal (a carbonyl compound) gives an orange crystalline solid; the alcohol prop-2-en-1-ol does not.

An extra "CH_3CH_2CHO" is printed after the word "compounds." in the stem of this question.

23.4 CARBOXYLIC ACIDS

Q.38 [Short] **What are carboxylic acids? Give common examples with their sources.**

Carboxylic acids are organic compounds having the carboxyl functional group (–COOH) in their molecules; their general formula is R–COOH. Common examples are methanoic acid (formic acid, found in ants), ethanoic acid (acetic acid, found in vinegar) and butanoic acid (butyric acid, in rancid butter).

Q.39 [Short] **Why do lower carboxylic acids have high boiling points and dissolve in water?**

Lower carboxylic acids are generally liquids at room temperature with high boiling points, because of their ability to form hydrogen bonds; ethanoic acid exists as a dimer held by two hydrogen bonds. Due to hydrogen bonding with water, the lower acids, i.e. methanoic acid and ethanoic acid, are soluble in water.

Q.40 [Short] **What is glacial acetic acid?**

The melting point of ethanoic acid is 17 °C and its boiling point is 118 °C. Its melting point is close to room temperature (25 °C), so it solidifies in winter. The solid form of pure ethanoic acid is called **glacial acetic acid** because of its resemblance to glaciers.

Q.41 [Short] **What are fatty acids? Why are they insoluble in water?**

Long-chain aliphatic monocarboxylic acids are called **fatty acids**. Stearic acid, palmitic acid and oleic acid are found in vegetable oils in the form of triglycerides. They are insoluble in water because of their long hydrophobic alkyl chains. Palmitic acid ($C_{16}H_{32}O_2$) is found in palm oil in large quantities.

Q.42 [Long] **Describe the methods of preparation of carboxylic acids.**

a) Oxidation of primary alcohols and aldehydes: the starting material is refluxed with excess of either acidified $KMnO_4$ or acidified $K_2Cr_2O_7$.

$$CH_3–CH_2–OH + 2[O] \rightarrow CH_3–COOH + H_2O \ (K_2Cr_2O_7 \text{ / dil. } H_2SO_4, \text{ heat})$$

$$CH_3–CHO + [O] \rightarrow CH_3–COOH\ (K_2Cr_2O_7 / \text{dil. } H_2SO_4, \text{heat})$$

b) Hydrolysis of alkane nitriles: nitriles are directly hydrolysed to carboxylic acids in acidic medium. In alkaline hydrolysis they first give salts of carboxylic acids, which yield the acids on acidification.

$$CH_3–CN + 2H_2O + HCl \rightarrow CH_3–COOH + NH_4Cl$$
$$CH_3–CN + H_2O + NaOH \rightarrow CH_3–COONa + NH_3\uparrow$$
$$CH_3–COONa + HCl \rightarrow CH_3–COOH + NaCl$$

c) Hydrolysis of esters: when an ester is refluxed with dilute acid or dilute alkali, it is hydrolysed to the carboxylic acid or the carboxylate ion.

$$CH_3–COO–CH_2–CH_3 + H_2O_{(l)} \rightleftharpoons CH_3–COOH + CH_3–CH_2–OH\ (H_3O^+)$$
$$CH_3–COO–CH_2–CH_3 + NaOH_{(aq)} \rightarrow CH_3–COONa + CH_3–CH_2–OH$$
$$CH_3–COONa + HCl \rightarrow CH_3–COOH + NaCl$$

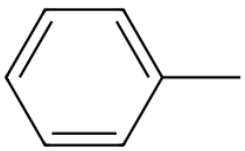
d) Preparation of benzenecarboxylic acid (benzoic acid): methylbenzene is readily oxidised by acidified $KMnO_4$ or $K_2Cr_2O_7$ to benzoic acid.

$$C_6H_5–CH_3 + 3[O] \rightarrow C_6H_5–COOH + H_2O\ (KMnO_4, \text{dil. } H_2SO_4, \text{heat})$$

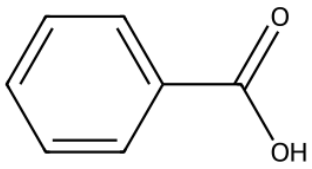
Q.43 [Short] How is benzoic acid prepared from methylbenzene?

Methylbenzene is heated with acidified $KMnO_4$ or $K_2Cr_2O_7$. The methyl side chain is oxidised to the –COOH group.

$$C_6H_5–CH_3 + 3[O] \rightarrow C_6H_5–COOH + H_2O$$



methylbenzene



benzoic acid (benzenecarboxylic acid)

Q.44 [Short] How does the hydrolysis of a nitrile differ in acidic and alkaline medium?

In acidic medium the nitrile is hydrolysed directly to the carboxylic acid (with an ammonium salt). In alkaline medium it first gives the salt of the carboxylic acid and ammonia; the acid is then obtained by acidification.

Quick Check 23.3

Q.45 [Short] **a) Write the equation for the oxidation of propanol with acidified $K_2Cr_2O_7$?**

$$CH_3–CH_2–CH_2–OH + 2[O] \rightarrow CH_3–CH_2–COOH + H_2O \ (K_2Cr_2O_7 \text{ / dil. } H_2SO_4, \text{ reflux})$$

Propan-1-ol is oxidised through propanal to propanoic acid.

Q.46 [Short] **b) What products you expect from following? Give the chemical equations. I) Hydrolysis of CH_3CH_2CN ii) Oxidation of ethylbenzene iii) Hydrolysis of ethyl ethanoate**

I) Propanoic acid (and ammonium chloride in acidic medium):

$$CH_3–CH_2–CN + 2H_2O + HCl \rightarrow CH_3–CH_2–COOH + NH_4Cl$$

ii) Benzoic acid; the whole side chain is oxidised to –COOH:

$$C_6H_5–CH_2–CH_3 + 6[O] \rightarrow C_6H_5–COOH + CO_2 + 2H_2O \ (KMnO_4, \text{ dil. } H_2SO_4, \text{ heat})$$

iii) Ethanoic acid and ethanol:

$$CH_3–COO–CH_2–CH_3 + H_2O \rightleftharpoons CH_3–COOH + CH_3–CH_2–OH \ (H_3O^+)$$

The textbook gives the oxidation of methylbenzene only; the ethylbenzene equation is written on the same pattern.

23.5 REACTIONS OF CARBOXYLIC ACIDS

Q.47 [Short] **Into which types are the reactions of carboxylic acids divided?**

Reactions of carboxylic acids are divided into three types:

1. Reactions involving the acidic hydrogen only.
2. Reactions involving the replacement of the –OH group.
3. Those involving the carboxyl group as a whole.

Q.48 [Long] **Describe the reactions of carboxylic acids involving the acidic hydrogen.**

Carboxylic acids show considerable acidic character because they give H^+ ions in water. That is why they react with basic species and active metals.

a) With active metals (a redox reaction): a salt and hydrogen are formed.

$$2CH_3–COOH + 2Na \rightarrow 2CH_3–COONa + H_2$$

b) With alkalis (neutralisation): a salt and water are formed.

$$CH_3–COOH + NaOH \rightarrow CH_3–COONa + H_2O$$

c) With metal carbonates and bicarbonates: a salt, water and carbon dioxide are formed.

$$2CH_3\text{–}COOH + Na_2CO_3 \rightarrow 2CH_3\text{–}COONa + CO_2 + H_2O$$
$$CH_3\text{–}COOH + NaHCO_3 \rightarrow CH_3\text{–}COONa + CO_2 + H_2O$$

Q.49 [Long] Describe the reactions of carboxylic acids in which the –OH group is replaced.

Carboxylic acids undergo substitution reactions in which the –OH group is replaced by some other group such as X, OR, NH_2, etc.

a) Reaction with $SOCl_2$, PCl_3 and PCl_5: acyl chlorides are formed.

$$CH_3\text{–}COOH + SOCl_2 \rightarrow CH_3\text{–}COCl + HCl + SO_2$$
$$CH_3\text{–}COOH + PCl_5 \rightarrow CH_3\text{–}COCl + HCl + POCl_3$$
$$3CH_3\text{–}COOH + PCl_3 \rightarrow 3CH_3\text{–}COCl + H_3PO_3 \text{ (heat)}$$

b) Reaction with alcohols (esterification): on heating with alcohols in the presence of concentrated sulphuric acid, carboxylic acids produce esters.

$$CH_3\text{–}COOH + C_2H_5OH \rightleftharpoons CH_3\text{–}COOC_2H_5 + H_2O \text{ (conc. } H_2SO_4)$$

The textbook prints the equations with $SOCl_2$ and PCl_5; the PCl_3 equation is written on the same pattern as for alcohols.

Q.50 [Short] How are carboxylic acids reduced? Which reducing agents cannot be used?

The powerful reducing agent lithium tetrahydridoaluminate(III), $LiAlH_4$, reduces carboxylic acids to primary alcohols. Other common reducing agents, H_2/Ni and sodium tetrahydridoborate(III), $NaBH_4$, will not reduce carboxylic acids.

$$CH_3\text{–}COOH + 4[H] \rightarrow CH_3\text{–}CH_2\text{–}OH + H_2O \text{ } (LiAlH_4)$$

Q.51 [Short] Define oxidation and reduction in five ways.

Oxidation	Reduction
Loss of H_2 (or H^-)	Gain of H_2 (or H^-)
Addition of O or O_2	Loss of O or O_2
Addition of X_2 (halogens)	Loss of X_2 (halogens)
Loss of electrons	Gain of electrons
Increase in oxidation state	Decrease in oxidation state

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Q.52 [Long] Why is methanoic acid a strong reducing agent? How is it identified?

Methanoic acid gets oxidised by both weak and strong oxidising agents because of its structure, and hence acts as a strong reducing agent. Its molecule, H–CO–OH, contains an **aldehydic portion (–CHO)**, responsible for its behaviour as a powerful reducing agent, and a **carboxylic portion (–COOH)**, responsible for its acidic character.

It is the only monocarboxylic acid which gives a positive silver mirror test (Tollens' test) and a brick-red precipitate with Fehling's solution. These reactions are used for the identification of methanoic acid.

$$HCOOH + 2[Ag(NH_3)_2]OH \rightarrow 2Ag\downarrow + CO_2 + 4NH_3 + 2H_2O$$

$$HCOOH + 2Cu(OH)_2 \rightarrow Cu_2O\downarrow + CO_2 + 3H_2O \text{ (brick-red ppt)}$$

Q.53 [Short] What happens when ethanedioic acid is treated with warm acidified potassium manganate(VII)?

Ethanedioic acid, HOOC–COOH, is oxidised by warm acidified $KMnO_4$ or $K_2Cr_2O_7$ to carbon dioxide and water.

$$2KMnO_4 + 3H_2SO_4 + 5(COOH)_2 \rightarrow K_2SO_4 + 2MnSO_4 + 10CO_2 + 8H_2O$$

23.6 RELATIVE ACIDITIES OF CARBOXYLIC ACIDS

Q.54 [Long] Compare the acidities of carboxylic acids, phenols and alcohols. Why are carboxylic acids the strongest?

Carboxylic acids are considerably more acidic than phenols, and phenols are more acidic than alcohols. The acidic strength, i.e. the extent of ionisation, depends on the stability of the conjugate base.

Substance	K_a	pK_a
RCOOH	10^{-5}	5
ArOH	10^{-10}	10
HOH	10^{-14}	14
R–OH	10^{-16}	16

$$CH_3\text{–}COOH \rightleftharpoons CH_3\text{–}COO^- + H^+ \; K_a = 1.8 \times 10^{-5}, \; pK_a = 4.76$$

In the ethanoate ion (the conjugate base), the negative charge on oxygen is decreased by delocalisation of electrons around the $–COO^-$ group (two equivalent resonance structures). The ion is stabilised and is less likely to accept H^+ ions to re-form the undissociated acid.

Q.55 [Short] What is pKa? How is it related to acid strength?

$pK_a = -\log K_a$. The higher the K_a, or the lower the pK_a, the stronger is the acid, and vice versa.

Q.56 [Short] How do electron-donating and electron-withdrawing groups affect the strength of a carboxylic acid?

An electron-donating group (+I effect) destabilises the carboxylate ion by intensifying its negative charge, and so decreases the acidity. An electron-withdrawing group (–I effect) stabilises the carboxylate ion by dispersing its negative charge, and so increases the acidity. For example, methanoic acid ($K_a = 1.8 \times 10^{-4}$, $pK_a = 3.75$) is about 10 times stronger than ethanoic acid ($K_a = 1.8 \times 10^{-5}$, $pK_a = 4.75$), because the methyl group of ethanoic acid has a +I effect.

Q.57 [Long] Explain the relative acidities of chlorine-substituted carboxylic acids.

Chlorine is an electron-withdrawing atom (–I effect). It pulls electron density away from the carboxylate group, disperses the negative charge and stabilises the ion, so the acid ionises more. The more chlorine atoms present, the stronger the acid. Trichloroacetic acid is comparable in strength to some mineral acids.

Acid	Formula	pK_a
Acetic (ethanoic) acid	CH_3–COOH	4.74
Chloroacetic acid	$ClCH_2$–COOH	2.86
Dichloroethanoic acid	Cl_2CH–COOH	1.26
Trichloroacetic acid	Cl_3C–COOH	0.64

Acid strength: Cl_3C–COOH > Cl_2CH–COOH > $ClCH_2$–COOH > CH_3–COOH

The textbook prints the pK_a of ethanoic acid as 4.76, 4.75 and 4.74 in different places, and writes "(–1 effect)" and "(+1 effect)" for the –I and +I effects.

Quick Check 23.4

Q.58 [Short] a) Methanoic acid is stronger than ethanoic acid. Give reason.

In ethanoic acid the methyl group is electron-donating (+I effect). It intensifies the negative charge on the ethanoate ion and destabilises it, which decreases the acidity. Methanoic acid has only a hydrogen atom in place of the methyl group, so its ion is more stable and it is about 10 times stronger.

Q.59 [Short] b) Place the following acids in order of strength, starting with the strongest acid first. i) CH_3CH_2COOH ii) CH_3CCl_2COOH iii) $CH_3CHClCOOH$

CH_3CCl_2COOH (ii) > $CH_3CHClCOOH$ (iii) > CH_3CH_2COOH (i)

Chlorine is electron-withdrawing (–I effect) and stabilises the carboxylate ion; two chlorine atoms have a greater effect than one.

Q.60 [Short] **c) Explain why ethanoic acid is a stronger acid than ethanol.**

In the ethanoate ion the negative charge is delocalised over the two oxygen atoms of the carboxylate group, so the ion is stabilised and is less likely to accept a proton again. In the ethoxide ion the charge stays on one oxygen atom and is intensified by the electron-donating ethyl group, so it is unstable. Hence ethanoic acid ionises much more than ethanol.

23.7 ESTERS

Q.61 [Short] **What are esters? Why are their boiling points and solubility lower than those of alcohols and acids? Give their uses.**

Esters are derivatives of carboxylic acids obtained by their reaction with alcohols. Unlike alcohols and carboxylic acids, esters cannot form hydrogen bonds with each other; this is why their boiling points are lower than those of the corresponding alcohols and carboxylic acids. The boiling point of ethyl ethanoate ($CH_3COOC_2H_5$) is 77 °C. The lack of hydrogen bonds also makes esters less soluble or insoluble in water. Many esters are volatile liquids with sweet fruity smells; they are used as solvents and as flavouring agents in fruit drinks and sweets. The fragrance and taste of fruits are due to esters.

Q.62 [Long] **Describe the methods of preparation of esters.**

i. By the reaction of a carboxylic acid and an alcohol: the reaction in acidic medium gives an ester. It is known as Fischer's esterification. The acid loses its –OH group and the alcohol loses the hydrogen of its O–H group.

$$CH_3–CH_2–COOH + CH_3–CH_2–OH \rightleftharpoons CH_3–CH_2–COO–CH_2–CH_3 + H_2O\ (H^+)$$

$$R–COOH + H–O–R \rightleftharpoons R–COOR + H_2O\ (H^+)$$

iii. By the reaction of acyl chlorides and alcohols: acyl chlorides react vigorously with alcohols to form esters. These reactions go to completion and do not form an equilibrium mixture like the reaction between carboxylic acids and alcohols.

$$CH_3–COCl + CH_3–CH_2–OH \rightarrow CH_3–COO–CH_2–CH_3 + HCl$$

iv. By the reaction of acyl chlorides and phenol: phenyl esters are formed when an acyl chloride is warmed with phenol (already discussed in Chapter 22).

$$C_6H_5–COCl + C_6H_5–OH \rightarrow C_6H_5–COO–C_6H_5\ \text{(phenyl benzoate)} + HCl$$

The methods are numbered i, iii and iv in the textbook (there is no ii). The phenyl benzoate equation is named in the learning outcomes and written on the pattern of phenyl ethanoate (Chapter 22).

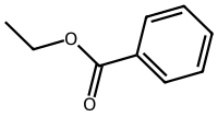
Q.63 [Short] **Why does the reaction of an acyl chloride with an alcohol give a better yield of ester than a carboxylic acid?**

The reaction of a carboxylic acid with an alcohol is reversible and forms an equilibrium mixture. Acyl chlorides react vigorously with alcohols and the reaction goes to completion.

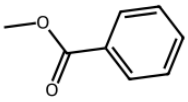
Q.64 [Long] Describe the hydrolysis of esters, including aromatic esters.

An ester is hydrolysed slowly to a carboxylic acid and an alcohol (or a phenol). The reaction is either acid- or base-catalysed. Aromatic esters are hydrolysed in the same manner, giving aryl carboxylic acids and alcohols or phenol.

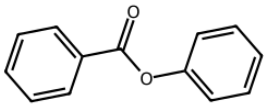
$$C_6H_5\text{–}COO\text{–}C_2H_5 + H_2O \rightarrow C_6H_5\text{–}COOH + C_2H_5OH\ (H^+/OH^-)$$
$$C_6H_5\text{–}COO\text{–}CH_3 + H_2O \rightarrow C_6H_5\text{–}COOH + CH_3OH\ (H^+/OH^-)$$



ethyl benzoate



methyl benzoate



phenyl benzoate

23.8 – 23.9 ACYL CHLORIDES AND HYDROLYSIS

Q.65 [Short] What are acyl chlorides? How are they named?

Acyl chlorides have the functional group –COCl. Saturated aliphatic acyl chlorides have the general formula $C_nH_{2n+1}COCl$. In the IUPAC system they are named as alkanoyl halides, e.g. ethanoyl chloride, CH_3–COCl, and propanoyl chloride, CH_3–CH_2–COCl. Acyl chlorides are the most reactive compounds among the functional derivatives of carboxylic acids, so they are usually prepared so that they can be used for the preparation of other functional derivatives.

Q.66 [Short] How are acyl chlorides prepared from carboxylic acids?

By the reaction of the carboxylic acid with $SOCl_2$, PCl_5, or PCl_3 and heat; the –OH group is replaced by Cl.

$$CH_3\text{–}COOH + SOCl_2 \rightarrow CH_3\text{–}COCl + HCl + SO_2$$

Q.67 [Long] Compare the ease of hydrolysis of acyl chlorides, alkyl chlorides and aryl chlorides.

The reactivity of acyl chlorides, alkyl chlorides and aryl chlorides can be compared by the ease of their hydrolysis.

$$CH_3\text{–}COCl + H_2O \rightarrow CH_3\text{–}COOH + HCl\ \text{(room temperature, rapid)}$$
$$CH_3\text{–}CH_2\text{–}Cl + H_2O \rightarrow CH_3\text{–}CH_2\text{–}OH + HCl\ \text{(heat under reflux)}$$
$$C_6H_5\text{–}Cl + H_2O \rightarrow \text{no reaction}$$

Compound	Reason	Ease of hydrolysis
Ethanoyl chloride (acyl chloride)	The carbonyl carbon has a large partial positive charge due to the strong electron-withdrawing effect of two highly electronegative atoms, oxygen and chlorine, attached directly to it. Nucleophilic attack of H_2O is very fast.	Complete within a few minutes at room temperature
Chloroethane (alkyl chloride)	The carbon gets a comparatively smaller partial positive charge because it is attached to only one chlorine atom. Nucleophilic attack of water is slow.	Takes 14 days even at 100 °C
Chlorobenzene (aryl chloride)	The lone pair on chlorine is delocalised with the π electrons of the benzene ring. This gives double-bond character to the C–Cl bond, making it so strong that it does not break.	No reaction

Ease of hydrolysis: acyl chloride > alkyl chloride > aryl chloride

Q.68 [Short] What is the addition–elimination mechanism of acyl chlorides?

Acyl chlorides react with nucleophiles such as water, alcohols and phenol by an **addition–elimination** mechanism. In the addition step, the lone pair of the nucleophile attacks the strongly δ+ carbonyl carbon and the π bond breaks, giving an intermediate with a negative charge on oxygen. In the elimination step, the C=O bond re-forms, a chloride ion is lost, and a proton is removed, giving HCl. Overall the Cl atom is replaced by the nucleophile.

The mechanism is named in the learning outcomes and in exercise MCQ X, but it is not drawn in the chapter; this is the standard description.

Quick Check 23.5

Q.69 [Short] a) Why do phenols not react directly with carboxylic acids to form phenyl esters?

In phenol a lone pair of the oxygen atom is delocalised into the benzene ring, so phenol is a poor nucleophile and cannot attack the less reactive carbonyl carbon of a carboxylic acid. Phenyl esters are therefore made from the much more reactive acyl chlorides.

This reason is not stated directly in the textbook; it follows from the delocalisation of the oxygen lone pair in phenol (Chapter 22) and the high reactivity of acyl chlorides.

Q.70 [Short] b) Phenyl ethanoate is the ester responsible for the fragrance of peach and strawberry. It can be prepared from the ethanoyl chloride and phenol. Write down a chemical reaction for the preparation of this ester.

$$CH_3\text{–}COCl + C_6H_5\text{–}OH \rightarrow CH_3\text{–}COO\text{–}C_6H_5 + HCl$$

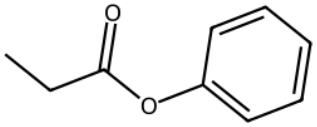
Q.71 [Short] **c) Predict the products of the following reactions, write the chemical equations. I) Chlorobenzene and ethanol ii) Propanoyl chloride and phenol**

I) No reaction. The C–Cl bond of chlorobenzene has double-bond character and is too strong to be broken by a nucleophile.

$$C_6H_5\text{–Cl} + C_2H_5\text{–OH} \rightarrow \text{no reaction}$$

ii) Phenyl propanoate and hydrogen chloride:

$$CH_3\text{–}CH_2\text{–COCl} + C_6H_5\text{–OH} \rightarrow CH_3\text{–}CH_2\text{–COO–}C_6H_5 + HCl$$



phenyl propanoate

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